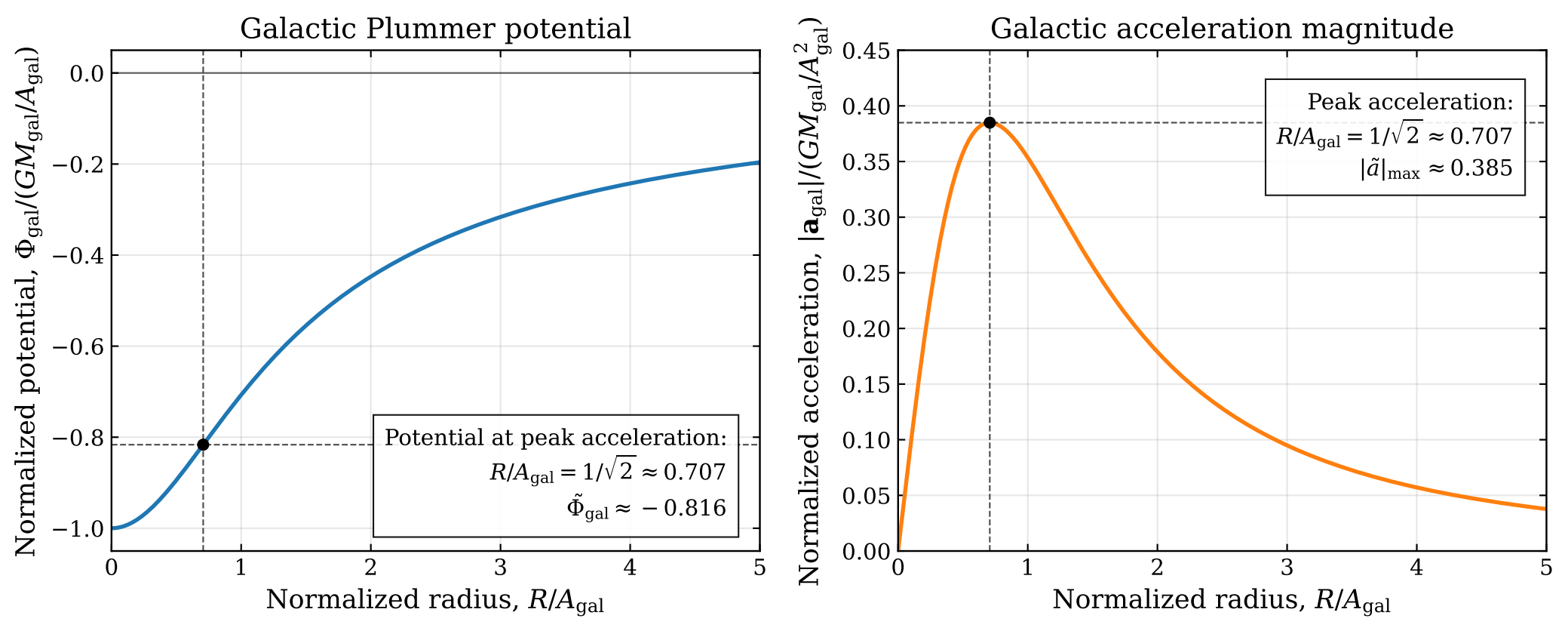
Galactic Plummer potential
Normalized potential, $\Phi_{\rm gal}/(GM_{\rm gal}/A_{\rm gal})$
Normalized radius, $R/A_{\rm gal}$
Potential at peak acceleration:
$R/A_{\rm gal} = 1/\sqrt{2} \approx 0.707$
$\tilde{\Phi}_{\rm gal} \approx -0.816$
Galactic acceleration magnitude
Normalized acceleration, $|\mathbf{a}_{\rm gal}|/(GM_{\rm gal}/A_{\rm gal}^2)$
Normalized radius, $R/A_{\rm gal}$
Peak acceleration:
$R/A_{\rm gal} = 1/\sqrt{2} \approx 0.707$
$|\tilde{a}|_{\rm max} \approx 0.385$